INCH-POUND
MIL-DTL-49285/18B
27 March 2018
SUPERSEDING
MIL DTL 49285/18A
31 March 2010

DETAIL SPECIFICATION SHEET

# CABLE, SPECIAL PURPOSE, ELECTRICAL, SIX INDIVIDUALLY SHIELDED PAIRS OF 22 AWG SOLID WIRE

Inactive for new design after 16 June 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-49285.

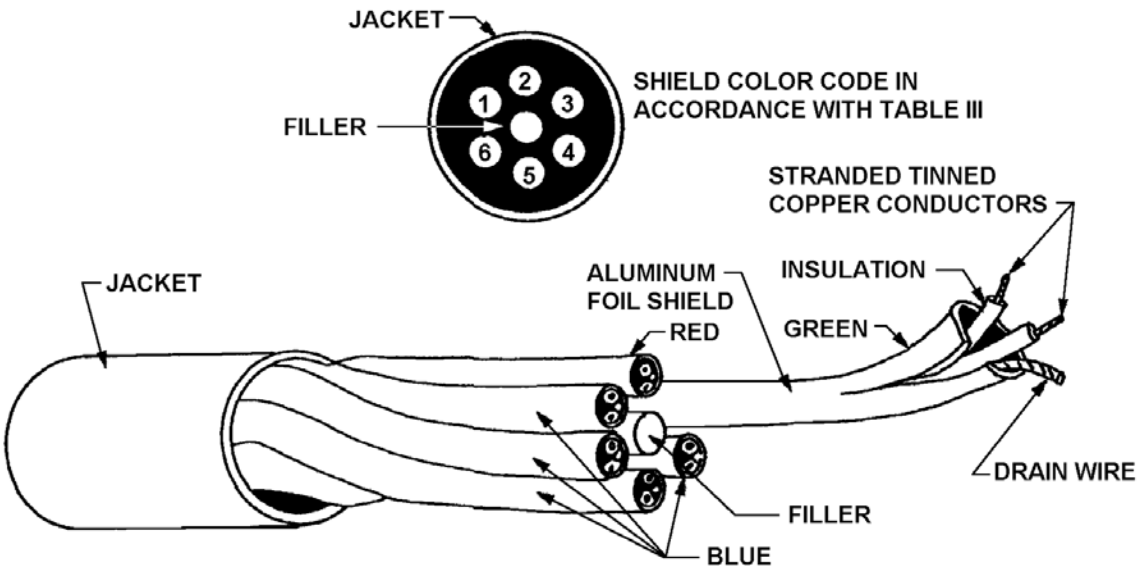



FIGURE 1. Cable illustration.

Engineering information: See table I.

Note: : Not recommended for shipboard or aerospace applications due to toxic smoke release in the event of a fire. PVC is not recommended in other applications (i.e. confined spaces, low ventilation) where there is the potential for human smoke exposure in a fire.



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TABLE I. Description, electrical.

| Electrical characteristics | Requirements |
|---|---|
| Nominal capacitance between conductors | 40 pF/ft (picofarads/foot) |
| Nominal capacitance between the conductor and the shield with one conductor tied to the shield | 77 pF/ft |
| Continuous working voltage | 300 V rms maximum between conductors |
| Nominal capacitance between adjacent shields | 150 pF/ft |
| Continuous working voltage between adjacent shields | 50 volts maximum |
| Shield dc resistance | 14 ohms/1,000 feet maximum |

TABLE II. Description, physical.

| Components | Construction details |
|---|---|
| Number of pairs | Six individually shielded |
| Conductor type and wire size | Solid tinned copper 22 AWG |
| Conductor insulation | Polyvinyl chloride |
| Conductor insulation thickness | 0.013 inch nominal thickness |
| Drain wire type and size | Solid tinned copper 22 AWG |
| Jacket material | Polyvinyl chloride |
| Jacket thickness | 0.035 inch |
| Finished cable diameter | 0.398 inch maximum outside diameter |
| Cable style (UL) | 2464 |
| Tensile strength (jacket) | 2,000 pounds per square inch minimum |
| Elongation (jacket) | 150 percent minimum |
| Overall cabling lay length | 1.3 twists per foot ± 10 percent |

Cabled pair orientation: The position of the red and green pair shield shall determine the rotation direction for pair location (see figure 1).



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TABLE III. Shield color code.

| Pair numbers | Shield color |
|---|---|
| Center core | |
| 1st pair | Red |
| 2nd pair | Green |
| 3rd pair | Blue – 1 |
| 4th pair | Blue – 2 |
| 5th pair | Blue – 3 |
| 6th pair | Blue – 4 |

REQUIREMENTS:

Design and construction: (see table II and table III):

Shield location and orientation. The polyester aluminum shield is to be located on the outer circumference of the pair, with the aluminum foil side inward. Insulation on the exterior of the shield is to be complete with no aluminum available for contact.

Drain wire location. The drain wire is to be spirally located beneath the shield with the insulated conductors. It is to be in continuous contact with the aluminum surface of the foil shield throughout the cable.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.

Flammability. The cable shall pass UL 2556 flame test requirements.

Shield integrity test.*

| Frequency | Response |
|---|---|
| 100 kHz | 114 dB or more below reference |
| 500 kHz | 105 dB or more below reference |
| 1 MHz | 99 dB or more below reference |
| 5 MHz | 85 dB or more below reference |
| 10 MHz | 79 dB or more below reference |
| 15 MHz | 74 dB or more below reference |
| 20 MHz | 70 dB or more below reference |
| 25 MHz | 65 dB or more below reference |
| 30 MHz | 61 dB or more below reference |
| 100 MHz | 42 dB or more below reference |

* Any two readings may deviate to a value of 0.9 times the stated limits.

Fixture resonance will occur between 40 and 70 MHz.



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Crosstalk test limits. All values are minimums.

| Frequency (kHz) | Near end crosstalk (dB) | Far end crosstalk (dB) |
|---|---|---|
| 40 | 58 | 58 |
| 70 | 56 | 55 |
| 100 | 54 | 53 |
| 1,000 | 55 | 52 |

Insulation resistance between shields: 100 megohms-1,000 feet minimum.

Durometer hardness. The cable jacket shall have a “Shore A” hardness of 82 ± 5.

Part or Identifying Number (PIN): The PIN shall be M49285/18.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. This document references MIL-DTL-49285 and UL 2556.

## CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army – MI
Air Force – 99
DLA – CQ

(Project 6145-2018-026)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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